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Signal Twin Review

A practical review sheet for understanding a legacy function, defining its target twin, and preparing a supportable release.

FUNCTION IDENTITY

Field	Review prompt	Record
Business function	What production decision or action does this function enable?	Function, process, line, product, and business consequence
Live users	Who uses, supervises, supports, or depends on it?	Operators, supervisors, quality, maintenance, engineering, R&D, support
Source signals	What enters the function?	PLCs, devices, tags, manual inputs, schedules, recipes, external systems
Logic	What behavior is encoded?	States, calculations, queries, scripts, transactions, exception handling
Consumers	Where does the output go?	Displays, reports, SQL jobs, PI points/AF, alerts, data lake, external apps
Consequence	What happens when the function is wrong, late, unavailable, or ambiguous?	Safety, quality, throughput, rework, downtime, release, support, customer impact
Ownership	Who accepts behavior, data, release, and ongoing support?	Named business, application, data, release, and support owners

TARGET TWIN DEFINITION

Layer	Target definition	Evidence to capture
Ignition application	Project/component boundaries, reuse, navigation, state, diagnostics, security	Architecture sketch, component inventory, state model, diagnostic behavior
Tag/UDT model	Naming, types, metadata, alarms, quality, ownership, versioning	Tag dictionary, UDT definition, quality and alarm rules, owner
SQL interaction	Named queries, transactions, schema, error handling, performance, access	Query map, schema contract, failure cases, access path, performance evidence
Historian context	Point mapping, asset hierarchy, units, timestamps, quality, retention, consumers	Consumer map, context model, stewardship, reversibility, retention rationale
Cloud contract	Schema, semantics, freshness, quality, access, owner, intended consumers	Data-product definition, acceptance criteria, named owner, consumer signoff

DEPENDENCY AND OWNERSHIP CHECK

Hidden dependencies

Queries, reports, scripts, PI consumers, spreadsheets, alerts, interfaces, and local workarounds.

Decision rights

Who can accept behavior, accept data, approve release, order rollback, and own support.

Change burden

Operator training, support readiness, procedure updates, data consumer changes, and communication.

REVIEWER NOTES

Primary risk

Notes:

Required evidence

Notes:

Named owner

Notes:

Release Decision

Promote, hold, repair, or retire based on explicit evidence and named authority.

ACCEPTANCE MATRIX

Check	Evidence required	Measure or observation	Status
Function parity	Normal states, transitions, calculations, commands, and exceptions	Matched tests, defects, recovery behavior	Pass / Hold / Repair
Data integrity	Names, values, units, timestamps, quality, completeness, reconciliation	Mismatch count, gaps, quality flags	Pass / Hold / Repair
Operator usability	Task flow, visibility, alarms, acknowledgment, recovery, training	Acceptance results, task friction, recovery clarity	Pass / Hold / Repair
Performance	Response, query/load behavior, store-forward, degradation handling	Latency, load, recovery, degraded-state behavior	Pass / Hold / Repair
Security and access	Roles, credentials, least privilege, network path, auditability	Access review, audit evidence, exception path	Pass / Hold / Repair
Supportability	Diagnostics, logs, runbook, owner, escalation, source/version	Diagnosis steps, handoff quality, owner readiness	Pass / Hold / Repair
Rollback	Trigger, authority, procedure, data reconciliation, communication	Rehearsal, elapsed time, decision clarity	Pass / Hold / Repair
Observation	Window, measures, owner, incident path, learning capture	Events, data quality, user behavior, incidents	Pass / Hold / Repair

DECISION RECORD

Promote

Evidence supports cutover; ownership, support, observation, and rollback are explicit.

Hold

Evidence is incomplete or production conditions, timing, or authority are not ready.

Repair

Target behavior, data context, usability, performance, security, or support needs change.

Retire

The legacy function is not worth reproducing; remove only with consumer and owner approval.

RELEASE RECORD

Decision and rationale

Promote / Hold / Repair / Retire:

Release and rollback authority

Names / roles:

Observation window

Dates / measures / owner:

LEARNING RETURN

Reusable component

What application, UDT, query, diagnostic, or test can be reused?

Reusable data rule

What naming, context, quality, stewardship, or cloud contract should become standard?

Reusable operating rule

What ownership, release, rollback, observation, or support practice should carry forward?

NEXT FUNCTION

Selection basis: operating value, dependency risk, representative architecture, reuse potential, data readiness, and production timing. Record the next candidate function and why it should follow.